<u>Unit 2 Biology AS</u>

Topic 3: Cell Structure, Reproduction and Development

3.1 Know that all living organisms are made of cells, sharing some common features

Key Concept:

- The **cell theory** states that:
  - All living organisms are composed of one or more cells.
  - The cell is the **basic structural and functional unit** of life.
  - All cells arise from pre-existing cells.

Shared Features of All Cells:

Feature	Function
Plasma membrane	Controls the movement of substances in and out of the cell
Cytoplasm	Site of metabolic reactions
DNA	Carries genetic information for controlling the cell
Ribosomes	Site of protein synthesis
Enzymes	Catalyse reactions essential to life
Energy release	All cells must carry out **respiration** (aerobic or anaerobic) to release energy

Even though structure varies, these core features exist in both **prokaryotic** and **eukaryotic** cells.

3.2 Understand how the cells of multicellular organisms are organised into tissues, tissues into organs, and organs into organ systems

Hierarchical Organisation in Multicellular Organisms:

1. **Cells**: The basic unit of life (e.g., epithelial cells, muscle cells).
2. **Tissues**: A group of similar cells working together to perform a specific function.
   - Example: Muscle tissue contracts; epithelial tissue lines surfaces.
3. **Organs**: Made of different tissues working together to carry out a function.
   - Example: The **heart** includes muscle tissue, nerve tissue, and connective tissue.
4. **Organ Systems**: Groups of organs that work together to carry out a broader function.
   - Example: **Circulatory system** = heart + blood vessels; **Digestive system** = stomach, intestines, liver, etc.

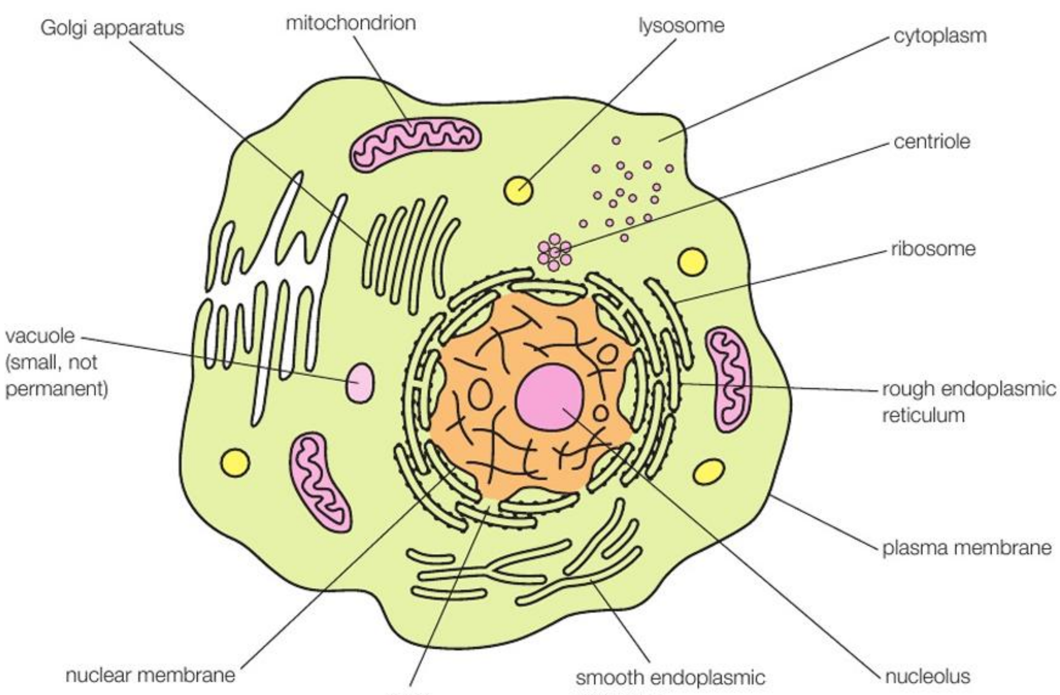
Specialisation:

- Multicellular organisms rely on **cell specialisation and cooperation** for survival, as no single cell can carry out all life functions.

3.3 Ultrastructure of Eukaryotic Cells

(i) Know the Ultrastructure:

Organelle	Structure	Function
Nucleus	Surrounded by nuclear envelope with pores; contains DNA and nucleolus	Stores genetic material; nucleolus makes rRNA and ribosomes
Nucleolus	Dense region inside nucleus	Synthesises ribosomal RNA
Ribosomes	Small, non-membranous; free or bound to rER	Site of protein synthesis
Rough Endoplasmic Reticulum (rER)	Membrane-bound sacs with ribosomes attached	Synthesises and processes proteins for secretion or membrane use
Smooth ER	Membrane-bound sacs without ribosomes	Lipid synthesis, detoxification
Mitochondria	Double membrane, inner folds = cristae, matrix inside	Site of aerobic respiration (ATP production)
Golgi Apparatus	Stacks of flattened sacs (cisternae), with vesicles	Modifies, packages, and transports proteins and lipids



Lysosomes	Vesicles containing hydrolytic enzymes	Digests old organelles, pathogens, and waste (intracellular digestion)
Centrioles	Cylindrical structures made of microtubules (in animal cells)	Involved in spindle formation during cell division

Note: Only animal and plant cells are eukaryotic. Plant cells also include chloroplasts, a large permanent vacuole, and a cellulose cell wall.

(ii) Understand Functions of Organelles:

- Protein synthesis and secretion involves coordination of **nucleus → rER → Golgi → vesicle → plasma membrane**.
- Organelles provide **compartmentalisation**, increasing efficiency of metabolic processes.
- Mitochondria generate energy, while lysosomes manage cellular waste.

3.4 Role of rER and Golgi in Protein Transport and Enzyme Secretion

Protein Transport Pathway (with modifications and refinement):

1. **Transcription in the nucleus** – A gene coding for the protein is transcribed into mRNA.
2. **mRNA exits the nucleus** – The mRNA travels through **nuclear pores** into the cytoplasm.
3. **Translation at ribosomes on the rough endoplasmic reticulum (rER)** – Ribosomes on the rER synthesize the **polypeptide chain** using the mRNA sequence.
4. **Protein enters rER lumen** – As the polypeptide is synthesized, it is threaded into the **lumen of the rER**.
5. **Protein folding and modification in the rER**:
    - The **polypeptide folds into its secondary and tertiary structures** with the help of chaperone proteins.
    - **Post-translational modifications** may occur, such as the addition of carbohydrates to form **glycoproteins**.
6. **Packaging into transport vesicles** – Properly folded and modified proteins are packaged into **transport vesicles** that bud off from the rER.
7. **Transport to the Golgi apparatus** – Vesicles fuse with the **cis face** of the Golgi apparatus.
8. **Further modification and sorting in the Golgi** – The Golgi may carry out additional modifications (e.g., **adding phosphate or sulfate groups**) and sorts proteins based on their final destination.
9. **Packaging into secretory vesicles** – Modified proteins are packaged into **secretory vesicles** at the **trans face** of the Golgi.
10. **Exocytosis** – Secretory vesicles move to the **plasma membrane**, fuse with it, and **release the proteins outside the cell** by exocytosis.

Formation of Extracellular Enzymes:

- **Digestive enzymes** (e.g., amylase, protease) are synthesised this way and **secreted out of the cell** to act on substrates in the gut. (mentioned last page)

3.5 Prokaryotic Cells

(i) Know the ultrastructure of prokaryotic cells, including:

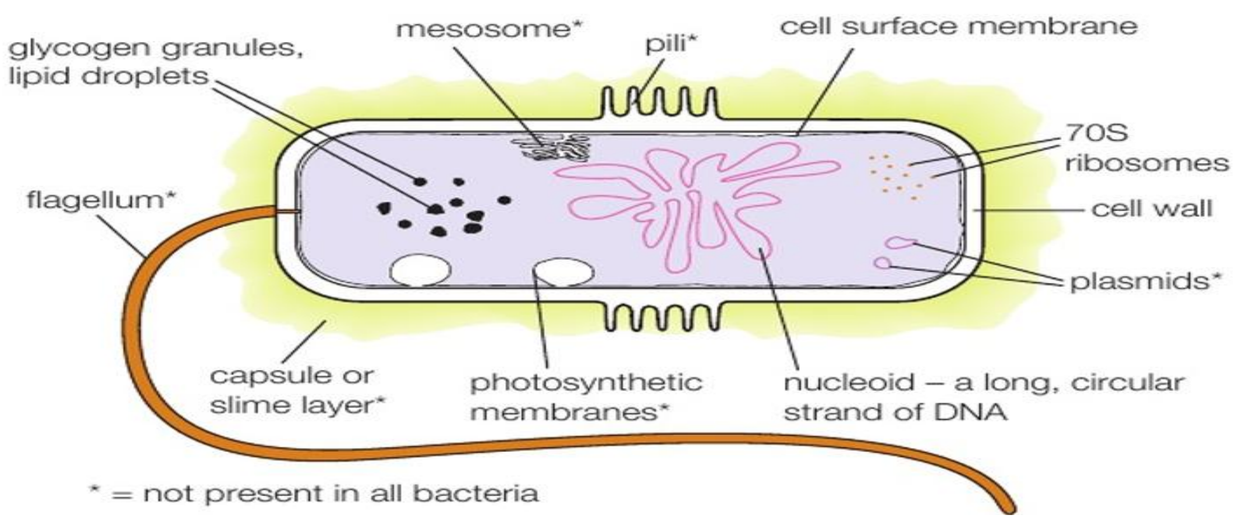
Structure	Description	Function
Cell wall	Made of **peptidoglycan (murein)**, not cellulose	Provides strength and prevents lysis
Capsule	Outer polysaccharide layer (slimy)	Protects from desiccation, immune system, and antibiotics
Plasmid	Small circular DNA molecule, separate from main DNA	Carries genes for antibiotic resistance and other traits
Flagellum	Long protein tail, rotates	Enables motility
Pili (fimbriae)	Short protein projections	Used for attachment or conjugation (DNA transfer)
Ribosomes	**70S type**, smaller than eukaryotic 80S	Site of protein synthesis
Circular DNA	Single loop of DNA not bound in a nucleus	Carries genetic information; floats free in cytoplasm

Key Notes:

- No **nucleus**, no **membrane-bound organelles**.
- Some prokaryotes are capable of forming **endospores** under stress.

(ii) Understand the function of the structures listed above:

- Prokaryotic cells are adapted for **survival in diverse environments**.
- Capsule aids in **pathogenicity** (e.g. in Streptococcus pneumoniae).
- Plasmids allow for **horizontal gene transfer** and rapid evolution.



3.6 Electron Micrograph Interpretation of Eukaryotic Organelles

You must be able to:

- **Recognise and label** organelles such as:
    - **Nucleus**, **nucleolus**
    - **Mitochondria**
    - **Golgi apparatus**
    - **rER and sER**
    - **Ribosomes**
    - **Lysosomes**
    - **Centrioles**

Key Identification Features:

Organelle	What to Look For in EM Image
Nucleus	Double membrane with pores; often largest structure
Nucleolus	Dense dark region within nucleus
Mitochondria	Double membrane, **cristae** (folded inner membrane)
rER	Flattened sacs with **dots (ribosomes)** attached
sER	Similar to rER but **no dots**
Golgi	Stacks of flattened sacs, often with vesicles nearby
Ribosomes	Tiny dots (more visible in clusters – polyribosomes)
Centrioles	Cylindrical bundles of microtubules, often in pairs

Lysosomes	Spherical vesicles, enzyme-filled (not easily distinguished unless labelled)

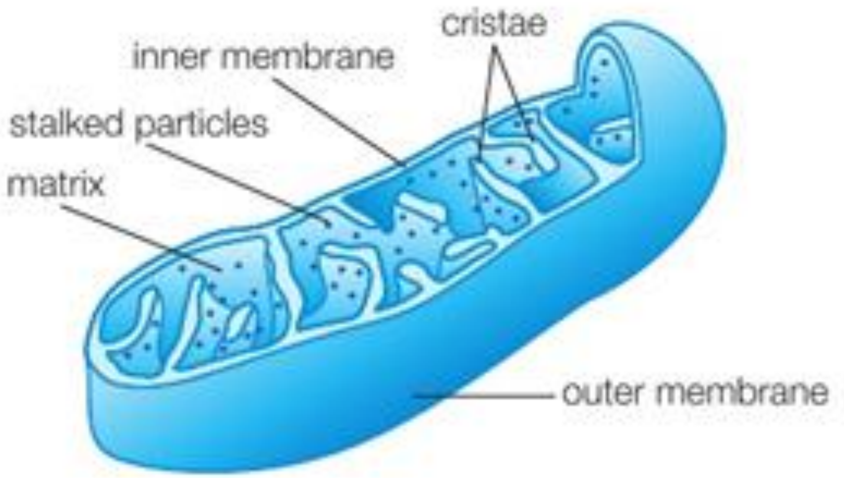


Fig: Mitochondria

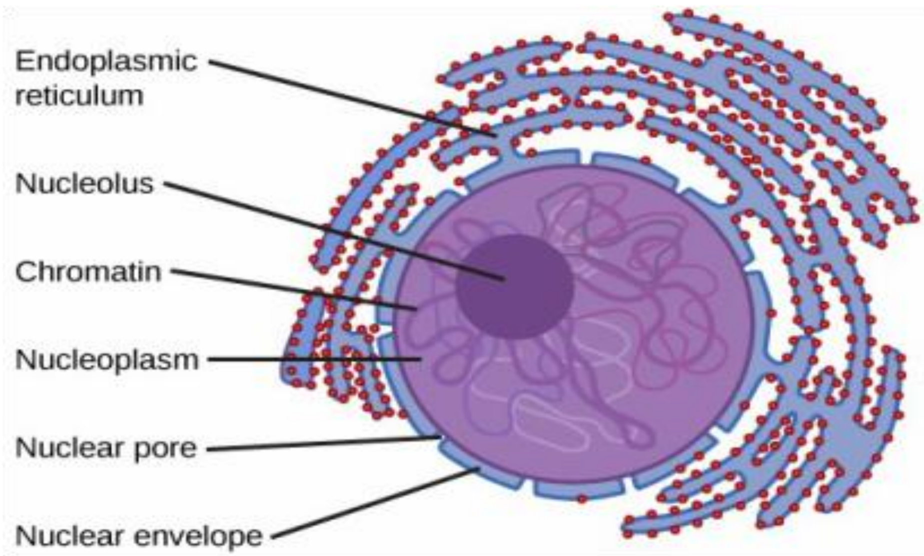


Fig: Nucleus and rER

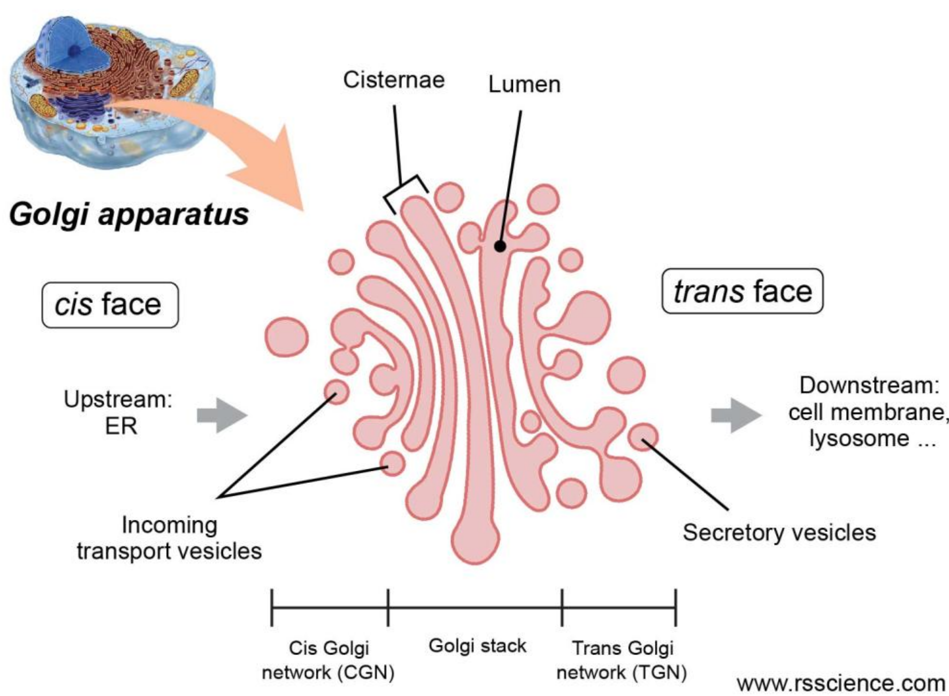


Fig: Golgi apparatus

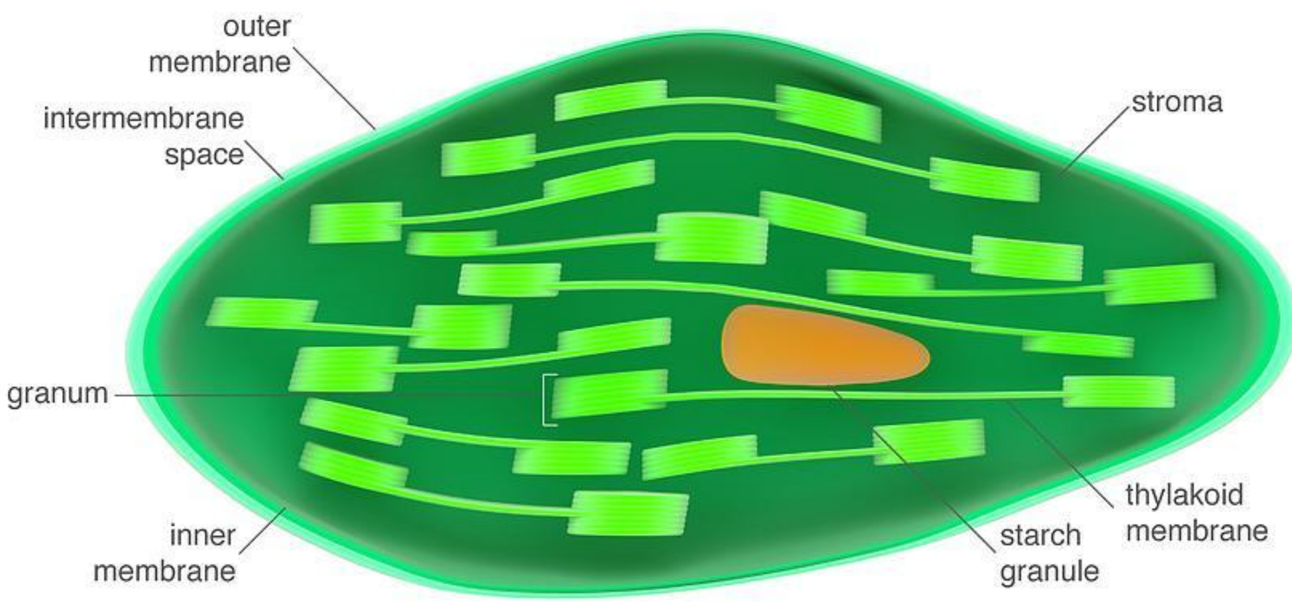


Fig: Chloroplast

3.7 Microscopy Principles

(i) Magnification and Resolution

Term	Definition
Magnification	How much larger the image is compared to the actual object
Resolution	The ability to **distinguish two close points** as separate

Calculating Magnification:

- Ensure **consistent units** (convert mm to µm etc.)
- **Image size** measured with a ruler
- **Actual size** can be found using a **scale bar** or graticule

$$\text{Magnification} = \frac{\text{Image size}}{\text{Actual size}}$$

(ii) Importance of Staining Specimens

- **Stains bind to specific cell components**, increasing contrast.
- Examples:
  - **Methylene blue**: binds DNA/RNA (nucleus)
  - **Iodine**: stains starch in plant cells
  - **Acetic orcein**: binds chromosomes in mitosis

Without staining, many structures would appear transparent and indistinguishable under the microscope.

3.8 Core Practical 5

(i) Use a light microscope to make observations and labelled drawings of suitable animal cells

Procedure:

- Collect animal cells (e.g. cheek epithelial cells).
- Place on slide with stain (e.g. methylene blue).
- Cover with coverslip carefully.
- Observe under **low then high power** objective lenses.

Drawing Guidelines:

- Clear lines
- No shading
- Label: nucleus, cytoplasm, cell membrane
- Use actual scale if possible

(ii) Use a graticule with a microscope to make measurements and understand scale

Eyepiece graticule:

- Transparent ruler fitted in the eyepiece

Stage micrometer:

- Used to **calibrate the graticule**
- Allows you to calculate the **length per graticule unit**

Once calibrated, you can:

- Measure size of cells or organelles in **micrometres (µm)**
- Use this for drawing scale bars

3.9 – Know that a locus is the location of genes on a chromosome

Definition:

- A **locus (plural: loci)** is the **specific, fixed position** on a chromosome where a particular **gene** is located.

Key Notes:

- Homologous chromosomes have the **same loci**, meaning they carry **genes for the same traits** — though they may carry **different alleles**.
- The **order of genes** is identical on homologous chromosomes.
- Knowing a gene's locus is crucial for:
    - **Mapping genes**
    - **Studying genetic linkage**
    - **Genetic screening and therapy**

3.10 – Understand the role of meiosis in ensuring genetic variation

Meiosis is a type of cell division that results in **four non-identical haploid gametes** from one diploid parent cell.

Genetic variation is introduced by:

(i) Independent Assortment (Metaphase I):

- Homologous chromosomes **line up randomly** at the equator.
- Each daughter cell gets a **random mix of maternal and paternal chromosomes**.
- This results in **2^n combinations**, where n = number of chromosome pairs.

(ii) Crossing Over (Prophase I):

- **Non-sister chromatids** of homologous chromosomes exchange segments as they join by chiasmata
- Produces **new combinations of alleles** on each chromatid (genetic recombination).
- Increases **genetic diversity** among gametes.

Random fertilisation

- Gametes being fertilised are random, so offspring characteristics will have random differences from the parents and therefore cannot be predicted with genotypic ratios in many cases

3.11 – Understand how mammalian gametes are specialised for their functions

Sperm Cell Adaptations:

Feature	Function
Acrosome	Contains enzymes to digest zona pellucida (acrosome reaction)
Nucleus	Haploid, carries one set of chromosomes
Tail	Enables motility to swim to egg
Mitochondria	Provide ATP for movement
Small size and streamlined	Easier to travel through female reproductive tract, reduced drag forces due to streamlined shape which increases speed of swimming and therefore fertilisation more likely to occur before sperm cell dies

Egg Cell Adaptations:

Feature	Function
Zona pellucida	Thick glycoprotein layer — protects egg; blocks polyspermy due to cortical reaction where it gets hardened
Cytoplasm	Rich in nutrients to support early development Such as lipid granules which can be burnt to release energy for embryo growth and mitosis
Haploid nucleus	Ensures correct chromosome number after fertilization as two haploid nuclei fuse to form a diploid nucleus (restoring the original chromosome number in zygote
Cortical granules	fuses with cell surface membrane of egg, releases enzymes by exocytosis during cortical reaction to harden zona pellucida
Large size	Accommodates early divisions and materials for embryo

3.12 – Fertilisation in Mammals

1. **Sperm reaches the egg** in the oviduct.
2. **Acrosome reaction**:
   - Enzymes from the sperm's acrosome digest the **zona pellucida**.
3. **Sperm membrane fuses** with egg membrane.
4. **Cortical reaction**:
   - Cortical granule fuses with egg cell's membrane
   - Enzymes are released by exocytosis into the zona pellucida

- These **harden the zona pellucida** to prevent polyspermy.

5. **Fusion of nuclei**:
    - Sperm and egg nuclei fuse to form a **diploid zygote**.

This process restores the **diploid number of chromosomes** and initiates development.

3.13 – Fertilisation in Flowering Plants

1. **Pollen grain lands** on the stigma (pollination).
2. **Pollen tube nucleus controls production and secretion of hydrolytic enzymes**
    - Hydrolytic enzymes digest the style
    - And creates a pathway for the generative nucleus which is called pollen tube
    - Grows through **micropyle** of ovule.
3. **Generative nucleus travel** down the tube:
    - Generative nucleus divides by mitosis to produce two male nuclei
    - One fertilises the egg cell → forms **diploid zygote**.
    - One fuses with polar nuclei → forms **triploid endosperm** (nutritive tissue).

This is known as **double fertilization**, unique to flowering plants.

Process	Mammals	Flowering Plants
Site	Oviduct	Ovule
Gametes	Sperm + egg	Pollen nucleus + egg
Polyspermy Prevention	Cortical reaction	Not needed (only one pollen tube reaches egg)
Zygote	Diploid	Diploid
Extra Product	–	Triploid endosperm

3.14 – Understand the Role of Mitosis and the Cell Cycle

The Cell Cycle Overview:

The cell cycle consists of **interphase** and **mitotic phase**.

1. **Interphase** (≈90% of cycle)
    - **G_1 phase**: Cell grows, material accumulation and protein synthesis (DNA polymerase enzymes etc)

- **S phase**: DNA replication → chromosomes double.
- **G_2 phase**: More growth, preparation for mitosis, organelle replication, more protein synthesis

2. **Mitotic Phase (M phase)**:
   - **Mitosis**: Division of the nucleus.
   - **Cytokinesis**: Division of the cytoplasm → two identical daughter cells.

Role of Mitosis:

- Produces **genetically identical diploid cells**.
- Essential for:
  - **Growth** of organisms
  - **Repair** of damaged tissues
  - **Asexual reproduction** (e.g., binary fission, budding)

Mitosis ensures that **daughter cells receive exact copies** of DNA, maintaining genetic stability.

3.15 – Recommended Additional Practical: Factors Affecting Pollen Tube Growth

- **Pollen grains** can be placed on nutrient agar or sucrose solutions of varying concentration.
- Growth of the **pollen tube** is measured under the microscope.
- Factors tested: **sugar concentration**, **temperature**, **pH**.
- Purpose: Understand conditions necessary for **successful fertilisation in plants**.

3.16 – Core Practical 6: Root Tip Mitosis

Aim: Observe and identify different stages of mitosis in plant root tip cells.

Procedure Summary:

1. Cut **1–2 cm** from the growing root tip (e.g., garlic or onion).
2. Treat with **HCl** to break down cell walls.
3. Stain using **acetic orcein** or **toluidine blue** to highlight chromosomes.
4. Squash gently using coverslip to spread cells into a single layer.
5. View under **high power** microscope, but find sample within range of view initially using low power objective lens

Stages of Mitosis to Identify:

Stage	Key Features
Prophase	Chromosomes condense and become visible; nuclear envelope breaks down
Metaphase	Chromosomes line up at equator, attached to spindle fibres
Anaphase	Sister chromatids are pulled apart to opposite poles
Telophase	New nuclear envelopes form; chromosomes decondense, nuckleolus reappears

Cytokinesis follows → division of cytoplasm

Drawing Guidelines:

- Clear, unshaded lines
- Accurate chromosome arrangement
- Include **labels**: stage name, chromosomes, spindle fibres if visible

3.17 – Stem Cells and Medical Use

(i) Key Terms:

Term	Definition
Stem cell	Undifferentiated cell capable of dividing and developing into different cell types
Totipotent	Can give rise to **any cell type**, including **placental** tissue i.e. extra-embryonic cells (e.g., zygote, morula)
Pluripotent	Can give rise to **all body cells**, but **not placental tissue (extra-embryonic cells)** (e.g., cells in blastocyst)
Morula	Solid ball of undifferentiated totipotent cells (early embryonic stage)
Blastocyst	Hollow structure with an inner cell mass of **pluripotent cells**

(ii) Stem Cells in Medicine – Scientific and Social Considerations:

Uses in Therapy:

- Treatment of degenerative conditions (e.g., Parkinson’s, diabetes, spinal injuries)
- Potential for **organ regeneration**
- Research into **cancer, ageing**, and **genetic disorders**

Ethical and Societal Considerations:

Pro	Con
Can save/improve lives	Involves destruction of human embryos
Reduces organ donor shortage	Raises moral/religious objections
May replace long-term medication	Fear of misuse or “designer babies”
Encourages scientific progress	Access may be unequal

Sources of Stem Cells:

- **Embryonic stem cells** (blastocyst): pluripotent
- **Adult stem cells** (bone marrow, cord blood): multipotent
- **Induced pluripotent stem cells (iPSCs)**: reprogrammed adult cells → pluripotent

iPSCs offer potential to **avoid ethical concerns** of embryonic stem cells.

Stages of embryonic development:

Morula: Formed by repeated mitosis cell division of the zygote

- All cells are totipotent and can differentiate to form all types of specialized cells
- Surrounded by zona pellucida
- Solid ball of cells

Blastocyst

- Surrounded by trophoblast, which forms the pkacental tissue
- There is an inner cell mass of pluripotent cells, which form the embryo
- Fluid filled cavity inside the blastocyst provides space for cell movement and growth
- There is a surrounding zona pellucida that breaks down just before implantation

3.18 – Cell Specialisation through Differential Gene Expression

Key Concept:

- All cells in an organism contain the **same DNA** (genome).
- However, only **certain genes are expressed** in each cell type → leads to **cell specialisation**.

Differential Gene Expression Process:

1. In undifferentiated cells (e.g., stem cells), all genes are available for transcription.
2. During specialisation, **some genes are activated**, while others are **switched off**.

3. Gene activation occurs by epigenetic modification (DNA or histone acetylation)
4. Genes are switched off by histone or DNA methylation
5. **Active genes** are transcribed → produce **mRNA** → translated into **proteins**.
6. The proteins carry out their functions in the cell, giving it characteristic properties and adaptations, therefore permanently modifying it
7. These proteins:
    - Determine cell structure (e.g., receptors, cytoskeleton)
    - Control specific cell functions (e.g., enzymes, signalling molecules)

The combination of **active proteins** defines the cell's type and function.

Example:

- A **muscle cell** expresses genes for actin and myosin.
- A **nerve cell** expresses genes for neurotransmitter proteins.

3.19 – Post-Transcriptional Gene Expression (Alternative Splicing)

Key Concept:

- A single gene can give rise to **multiple proteins** by altering the **mRNA** after it is transcribed.

How It Works:

1. Pre-mRNA is made during transcription.
2. **Introns** are removed; **exons** are spliced together by spliceosomes
3. **Different combinations of exons** can be joined → produces **different mature mRNA molecules**.
4. Active mRNA may be shorter as some exons are removed
5. Each mRNA is translated into a **different protein**.

This allows for **protein diversity** without increasing the number of genes.

Example:

- The gene for tropomyosin (in muscles) can produce different versions in **skeletal** and **smooth muscle** via alternative splicing.

3.20 – Genotype, Environment and Epigenetics

(i) Phenotype = Genotype + Environment

- **Genotype**: the genetic makeup (alleles an organism possesses).
- **Environment**: external factors that influence gene expression (e.g., diet, toxins, temperature).
- Some phenotypes are controlled **solely by genes** (e.g., blood group), others by **gene-environment interactions** (e.g., height, skin colour).
- Mutagens such as UV are some environmental factors that directly influence the genotype, which in turns changes the phenotype aswell, because mutations change the base sequence in the DNA

(ii) Epigenetics: DNA Methylation and Histone Modification

These are **heritable changes in gene expression** that **do not alter the DNA sequence**.

1. DNA Methylation:

- Addition of **methyl groups (CH_3)** to cytosine bases in DNA.
- Often occurs at **CpG sites**.
- Methylation **inhibits mRNA binding with the DNA, inhibiting transcription**→ **gene is silenced**.
- Example: Silencing of tumor suppressor genes in cancer.

2. Histone Modification:

- Histone proteins can be chemically modified (e.g., **acetylation**, **methylation**).
- Affects how tightly DNA is wrapped around histones:
    - **Acetylation** → DNA loosely wrapped and mRNA able to bind→ gene **activated**.
    - **Deacetylation/methylation** → DNA tightly wrapped so bases are not exposed for mRNA to bind→ gene **silenced**.

Epigenetic changes are **reversible** and may be influenced by environment (e.g., diet, stress, toxins and mutagens such as UV).

(iii) Inheritance of Epigenetic Marks:

- During **cell division**, epigenetic marks are passed on to **daughter cells**.
- This ensures **stable gene expression patterns** during development (e.g., liver cells remain liver cells).
- Some epigenetic changes may be passed on to **offspring**, contributing to **transgenerational effects**.

3.21 – Polygenic Inheritance and Continuous Variation

Monogenic vs Polygenic:

Feature	Monogenic Traits	Polygenic Traits
Controlled by	One gene	Many genes
Type of variation	Discontinuous (e.g., blood group)	Continuous (e.g., height, skin colour)
Phenotype categories	Distinct	Range of phenotypes
Graph shape	Bar chart	Bell-shaped curve (normal distribution)

Polygenic Inheritance:

- Each gene involved contributes **additively** to the trait.
- The **more dominant alleles**, the more intense the trait expression.
- Environmental factors further influence phenotype (e.g., nutrition affects height).
- The more dominant allele the person inherits, the more intense the traits
- Each allele has a different base sequence, so they code for different proteins and therefore have different effects on the phenotype

Continuous Variation:

- Traits show a **range of values**, not categories.
- No clear boundaries between phenotypes.
- Examples: height, skin tone, intelligence, milk yield in cows.

Errors in meiosis

1. Non-disjunction

Definition:

- Failure of **homologous chromosomes** (meiosis I) or sister chromatids (meiosis II) to separate properly**.**
- Results in **gametes with extra or missing chromosomes.**

When it Occurs:

Stage	What Fails?	Example
Meiosis I	Homologous chromosomes don't separate	Both chromosomes go to one cell
Meiosis II	Sister chromatids don't separate	One gamete gets two copies of a chromosome

2. Resulting Conditions (Aneuploidy)

- **Aneuploidy = abnormal number of chromosomes in gametes or zygotes.**

Condition	Chromosome	Karyotype	Phenotype
Down syndrome	Extra chromosome 21	47 chromosomes (trisomy 21)	Developmental delay, distinct facial features
Turner syndrome	Missing one X (females)	45, XO	Short stature, infertility
Klinefelter syndrome	Extra X in males	47, XXY	Feminised features, low fertility
Patau syndrome	Trisomy 13	47 chromosomes	Severe developmental disorders, early death

Tumor and cancer

- uncontrolled cell division due to mutations in genes that regulate the cell cycle leads to the formation of a tumour, which is an abnormal mass of cells.
- These mutations often occur in:
  - Proto-oncogenes (which become oncogenes)
  - Tumour suppressor genes

Key Genes Involved in Cancer

Gene Type	Normal Role	Effect of Mutation
Proto-oncogenes	Promote normal cell division when needed	Become oncogenes; promote uncontrolled division
Tumour suppressor genes	Inhibit cell division or promote apoptosis	Inactivated; damaged cells continue dividing

Types of Tumours

Tumour Type	Description	Growth	Spread	Risk
Benign Tumour	Non-cancerous	Slow	Localised (does not invade tissues)	Usually not life-threatening (but may press on organs)
Malignant Tumour	Cancerous	Rapid and uncontrolled	**Invades** nearby tissues and can **spread** (metastasis)	Dangerous and often life-threatening

As the tumor grows, the tumor cells may become malignant if they break away from primary tumor and invade surrounding tissues and metastasize which is cancer. Disruptions in apoptosis further contribute to cancer development by allowing defective cells to survive and proliferate.